RTL NOR Gate

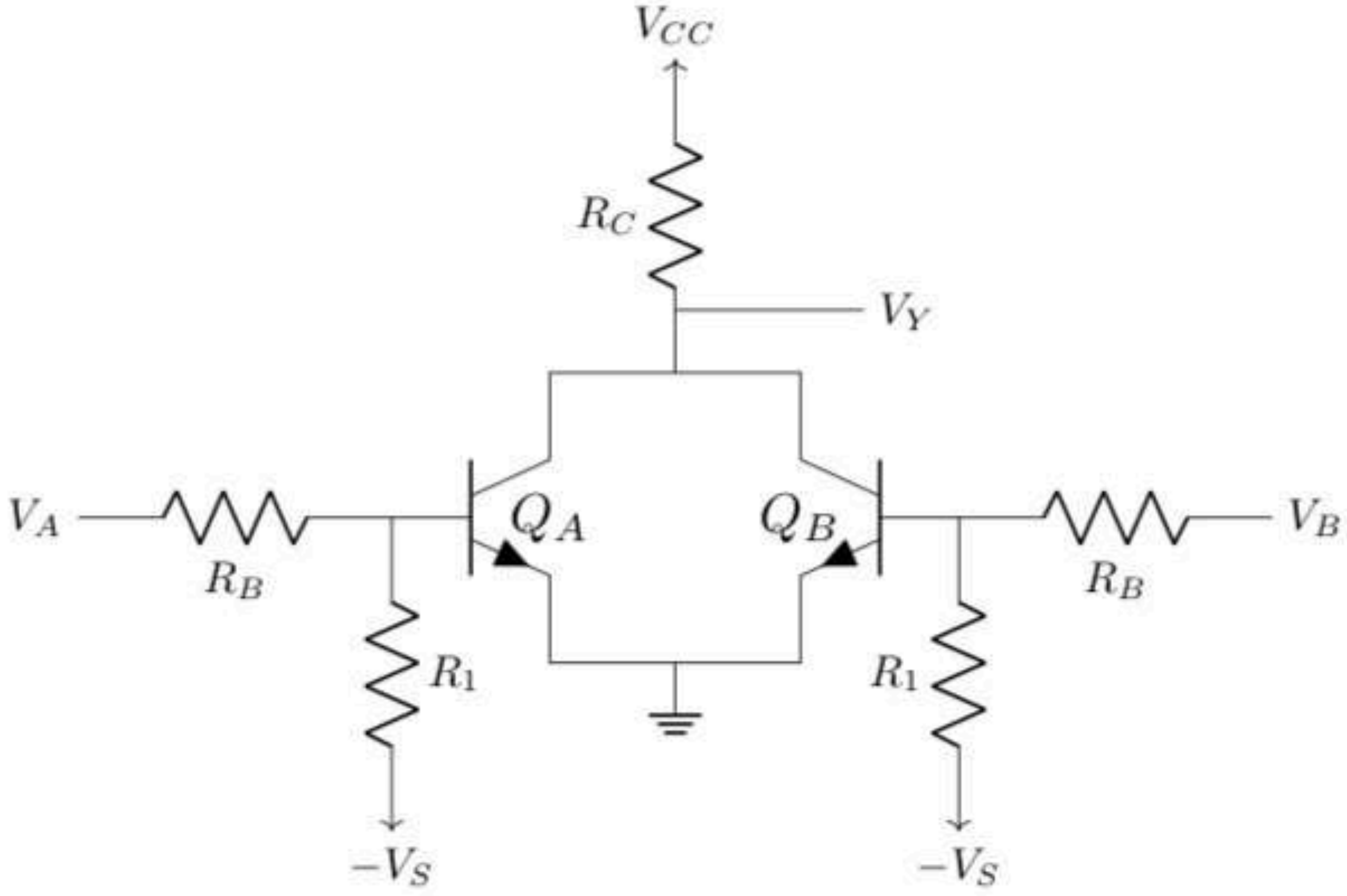


Figure 6.10: A typical RTL NOR gate

Although we have denoted the same value of R_B, V_S for the two BJTs, they can be different as well. Now that there are two inputs (V_A & V_B), there will be four input cases. Let's discuss them accordingly.

Case(0,0): When both the inputs are low, both Q_A & Q_B will be in cut-off. The result will be both of them acting as an open circuit. Then the source voltage V_{CC} appears directly at 'Y', as no current flows through R_C (as $I_{C_A} = I_{C_B} = 0$). So, $V_Y = V_{CC}$, which is a high voltage. This is the desired result for case (0,0) of a NOR gate.

Case(0,1) or (1,0): Since we have considered similar input side parameters (R_B & $-V_S$) for both the BJTs, case (1,0) & (0,1) would produce similar results. So, when one of the inputs is low, its BJT will be in cut-off and the other one's will be in saturation for the same reason as in the case of NOT gate. Now that one BJT is in saturation, its collector voltage (V_C) will be set to '0.2 V'. So, $V_Y = 0.2\,V$. Which is a low voltage. The collector current of this BJT (in saturation) will be non-zero and will depend on the external circuit (V_{CC} & R_C). Its base voltage (V_B) will be '0.8 V' too. So, the analysis will be similar to the NOT gate's case (1).

Case(1,1): When both inputs are high, both the BJTs will be in saturation. Then both their collector voltages will be set to '0.2 V'. So, again we get $V_Y = 0.2\,V$, a low output voltage. However, for this case, the current through R_C will split into two separate collector currents ($I_{R_C} = I_{C_A} + I_{C_B}$).

Let's look at the power dissipation related stuffs.

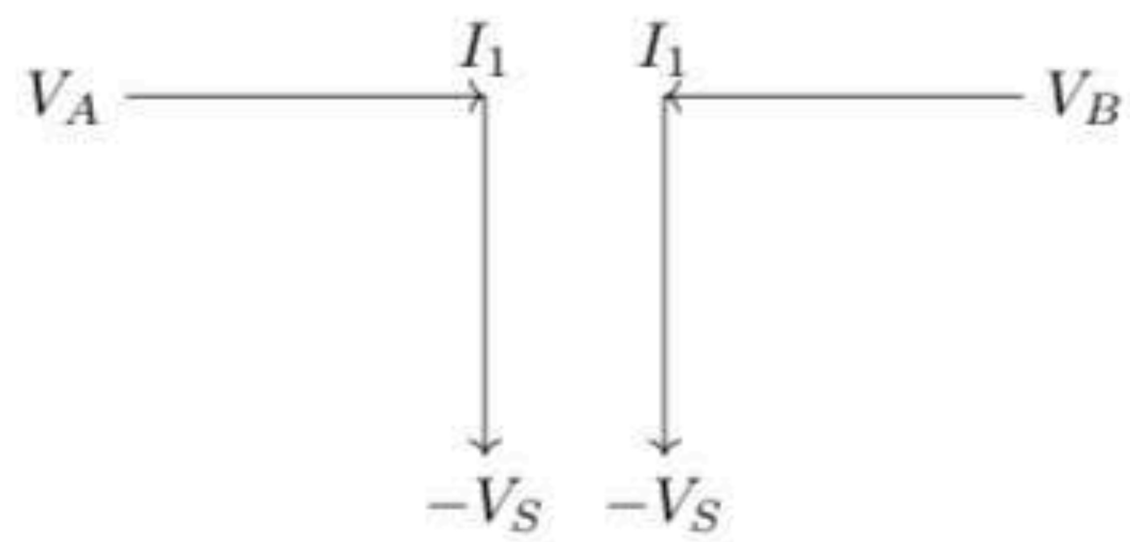


Figure 6.11: Power dissipation diagram for Case (0,0)

$P_{(0,0)} = 2 \times (V_A - (-V_S)) \times I_1$

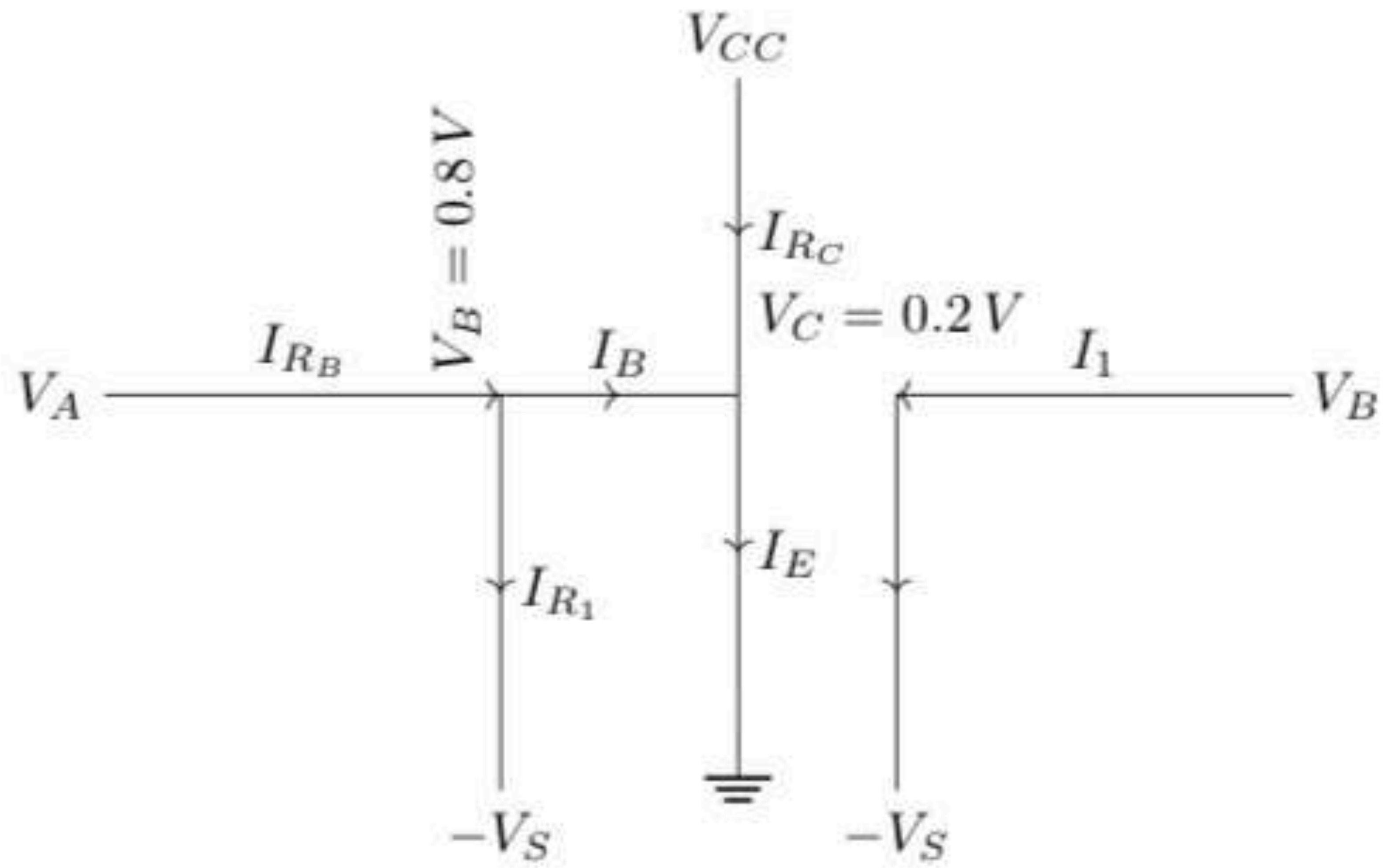


Figure 6.12: Power dissipation diagram for Case(1,0)

$P_{(1,0)} = (V_A - (-V_S)) \times I_{R_B} + (V_B - 0) \times I_B + (V_{CC} - 0) \times I_{R_C} + (V_B - (-V_S)) \times I_1$

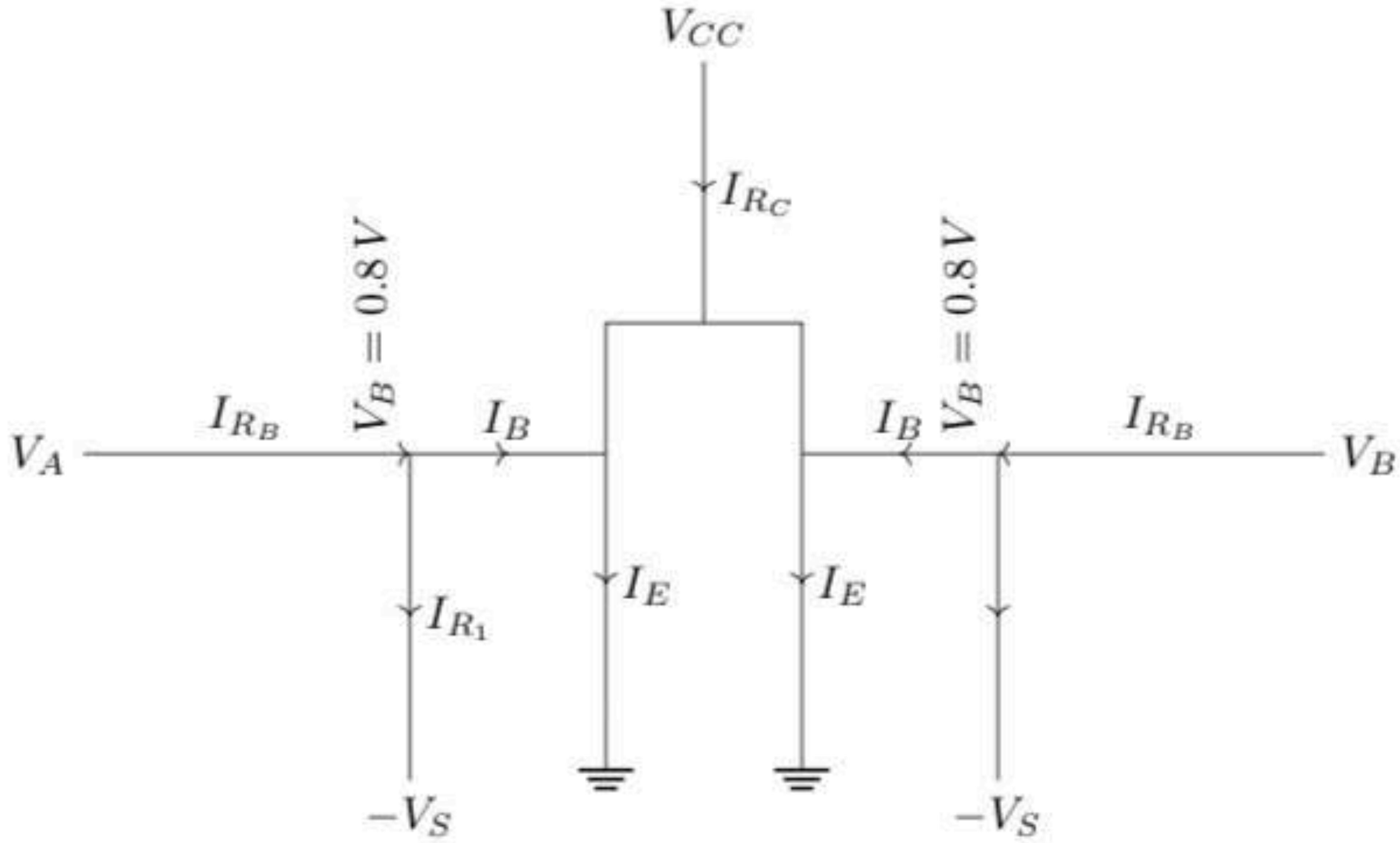
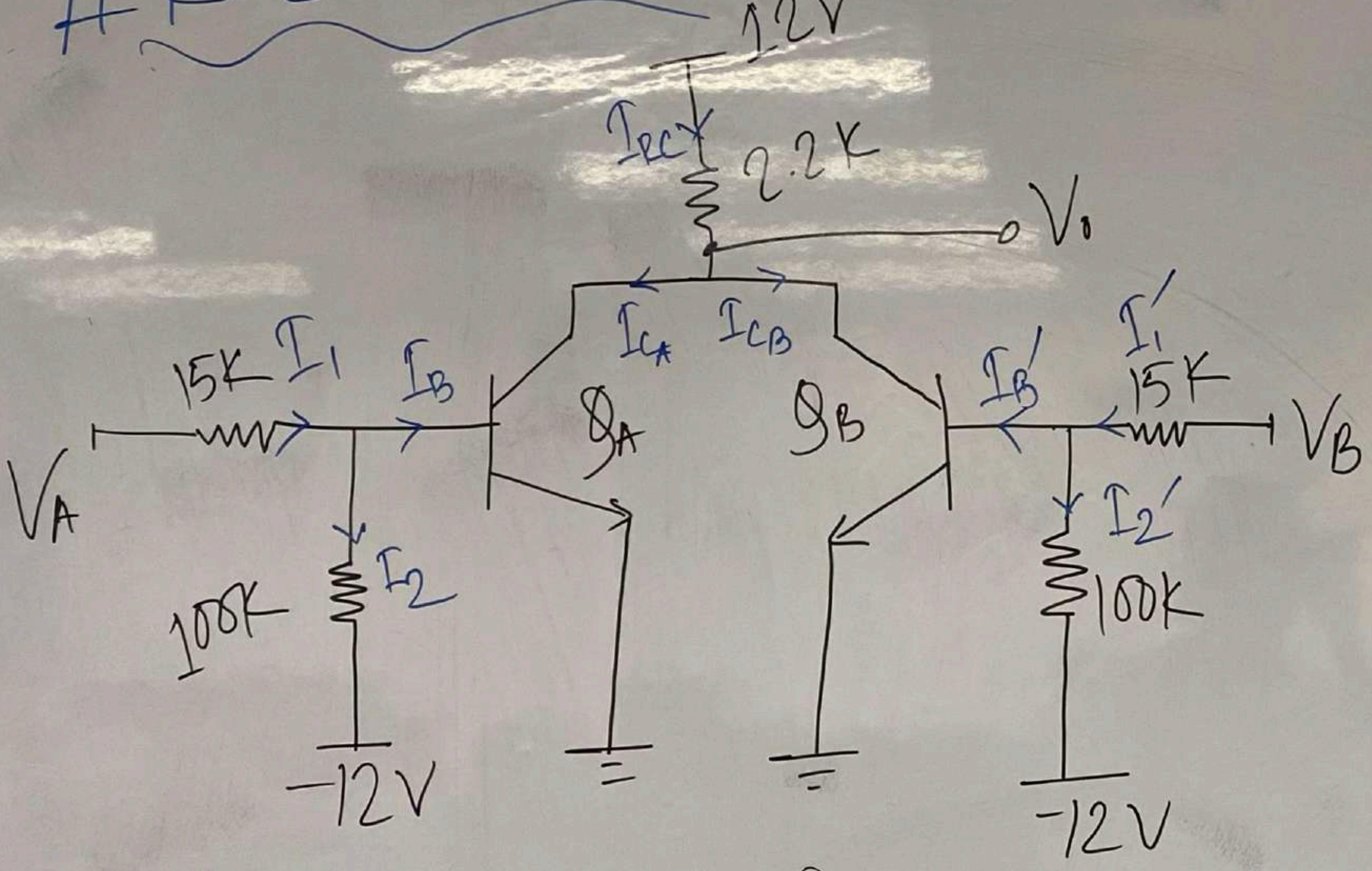


Figure 6.13: Power dissipation diagram for Case(1,1)

$P_{(1,1)} = 2 \times [(V_A - (-V_S)) \times I_{R_B} + (V_B - 0) \times I_B] + (V_{CC} - 0) \times I_{R_C}$

RTL NOR Gate



i) Find V_o and currents for all input cases

ii) Find Power Consumptions

iii) Noise Margin & Fanout → From Practice Sheets

Case - I

$V_A = 0V,\ V_B = 0V$

$Q_A \to$ Cutoff, $Q_B \to$ Cutoff

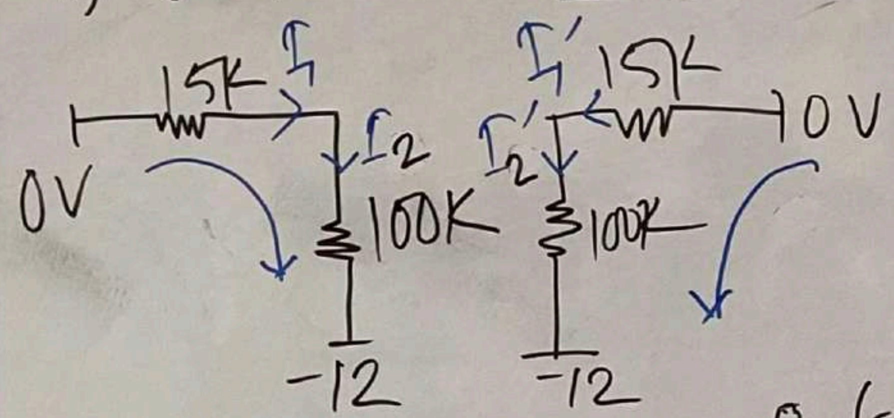
$\therefore I_{CA} = I_B = 0$

$I_{CB} = I_B' = 0$

$I_{RC} = I_{CA} + I_{CB} = 0$

$\therefore I_{RC} = \dfrac{12 - V_o}{2.2K}$

$\Rightarrow 0 = 12 - V_o \quad \therefore \boxed{V_o = 12V}$



here, $I_1 = I_2 = I_1' = I_2' = \dfrac{0-(-12)}{100+15}$ mA

Power, $P_1 = (0-(-12)) \times I_1 + (0-(-12)) \times I_1'$

$= 2 \times (0-(-12)) \times I_1$ mW

mA

$) \times I_1$

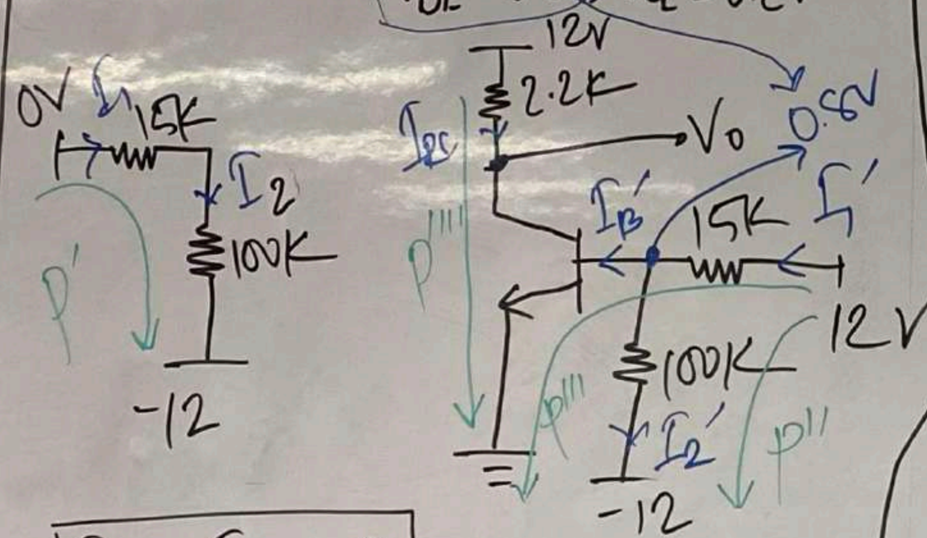
)

Case-II

$V_A = 0V,\ V_B = 12V$

$Q_A \rightarrow$ Cutoff, $Q_B \rightarrow$ Saturation

$\therefore V_{BE} = 0.8V,\ V_{CE} = 0.2V$



$\boxed{I_{CA} = I_B = 0}$

$I_1 = I_2 = \dfrac{0-(-12)}{100+15}\ mA$

$V_{CE} = 0.2 \rightarrow \therefore \boxed{V_0 = 0.2V}$

$I_{RC} = I_{CB} = \dfrac{12-0.2}{2.2K} =$

$I_1' = \dfrac{12-0.8}{15K} =$

$I_2' = \dfrac{0.8-(-12)}{100K} =$

$I_B' = I_1' - I_2'$

$P_2 = P' + P'' + P''' + P''''$

$P_2 = (0-(-12)) \times I_1 + (12-(-12)) \times I_2'$
$+ (12-0) \times I_B' + (12-0) \times I_{RC}$

Case-III

$V_A = 12V,\ V_B = 0V$

Same as Case-II

$\boxed{V_0 = 0.2V}$

$I_{CB} = I_B' = 0$

$I_1' = I_2' = \dfrac{12}{115}\ mA$

$I_{RC} = I_{CA} = \dfrac{12-0.2}{2.2}$

$I_1 = \dfrac{12-0.8}{15},\ I_2 = \dfrac{12.8}{100}$

$I_B = I_1 - I_2$

$P_3 = P_2$

Case-IV

$V_A = 12V,\ V_B = 12V$

$Q_A, Q_B \rightarrow$ both Saturation.

$V_{BE} = 0.8V,\ V_{CE} = 0.2V$

$\therefore \boxed{V_0 = 0.2V}$

$I_{RC} = \dfrac{12-0.2}{2.2K}\ mA,\quad \boxed{I_{CA} = I_{CB} = \dfrac{I_{RC}}{2}}$

$I_1 = I_1' = \dfrac{12-0.8}{15K}$

$I_2 = I_2' = \dfrac{0.8-(-12)}{100K}$

$I_B = I_B' = I_1 - I_2$

$P_4 = \{(12-(-12)) \times I_2 + (12-0) \times I_B + (12-0) I_{CA}\} \times 2$